

A finite-rank norm-attaining tensor with no finite optimal representation

arXiv automatic math run `fa_banach_001`, lane 10

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Status

This packet records a candidate counterexample, likely valid, to the finite-rank equality question asked in Aliaga–Dantas–Guerrero–Viu–Jung–Roldan [1]. The source question asks whether

$$\text{FNA}_\pi(X \widehat{\otimes}_\pi Y) = \text{NA}_\pi(X \widehat{\otimes}_\pi Y) \cap (X \otimes Y)$$

holds for arbitrary Banach spaces X, Y , beyond the easy cases listed in the source paper.

Source question

The question appears in Section 2.3 of [1], after the definition of finitely norm-attaining tensors. The relevant source crop is shown in Figure 1.

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2.3. Finitely norm-attaining tensors. Every integral norm-attaining tensor can be approximated by “classical” norm-attaining tensors. In fact, a stronger statement holds: such an approximation can be achieved by *finitely norm-attaining tensors*, that is, tensors $u \in X \widehat{\otimes}_\pi Y$ that admit a finite optimal representation of the form

$$u = \sum_{k=1}^n x_k \otimes y_k \quad \text{such that} \quad \sum_{k=1}^n \|x_k\| \|y_k\| = \|u\|_\pi.$$

Let us denote by $\text{FNA}_\pi(X \widehat{\otimes}_\pi Y)$ the set of finitely norm-attaining tensors in $X \widehat{\otimes}_\pi Y$.

Clearly all FNA_π tensors have finite rank and are norm-attaining. Both reverse implications fail: there are non-norm-attaining tensors in $L_1 \widehat{\otimes}_\pi \ell_2^2 = L_1 \otimes \ell_2^2$ [14, Example 3.12(a)], and every element of $\ell_1 \widehat{\otimes}_\pi Y$ is norm-attaining including those with infinite rank [14, Proposition 3.6]. On the other hand, we do not know whether the equality

$$\text{FNA}_\pi(X \widehat{\otimes}_\pi Y) = \text{NA}_\pi(X \widehat{\otimes}_\pi Y) \cap (X \otimes Y)$$

holds in general, outside of the easy cases where X, Y are finite-dimensional [14, Proposition 3.5] and where X is a finite-dimensional polyhedral space and Y is a dual space [13, Theorem 4.1].

Figure 1: Section 2.3 of arXiv:2602.23796, page 11, defining FNA_π and asking whether it equals $\text{NA}_\pi \cap (X \otimes Y)$ in general.

Definitions

An element $u \in X \widehat{\otimes}_\pi Y$ is in $\text{NA}_\pi(X \widehat{\otimes}_\pi Y)$ if it admits a countable optimal projective representation

$$u = \sum_{n=1}^{\infty} x_n \otimes y_n, \quad \|u\|_\pi = \sum_{n=1}^{\infty} \|x_n\| \|y_n\|.$$

It is in $\text{FNA}_\pi(X \widehat{\otimes}_\pi Y)$ if such an optimal representation exists with only finitely many summands.

Proof intuition

For $H = \ell_2^2$, the product $\ell_1 \widehat{\otimes}_\pi H$ is the vector-valued ℓ_1 -space $\ell_1(H)$, with norm $\sum_n \|v_n\|_H$. A tensor can therefore be norm-attaining by splitting it into its coordinate atoms. If a finite representation were also optimal, equality in the summed triangle inequality would have to occur at every coordinate. In a strictly convex two-dimensional Hilbert space, equality in the triangle inequality forces all nonzero summands at that coordinate to lie on the same line. A finite representation has only finitely many available lines, so a sequence whose coordinates point in infinitely many different directions cannot be finitely norm-attaining.

The counterexample

Theorem 1. *Let $H = \mathbb{R}^2$ with the Euclidean norm. There is an element*

$$u \in \text{NA}_\pi(\ell_1 \widehat{\otimes}_\pi H) \cap (\ell_1 \otimes H)$$

such that $u \notin \text{FNA}_\pi(\ell_1 \widehat{\otimes}_\pi H)$. Consequently,

$$\text{FNA}_\pi(X \widehat{\otimes}_\pi Y) = \text{NA}_\pi(X \widehat{\otimes}_\pi Y) \cap (X \otimes Y)$$

does not hold for all Banach spaces X, Y .

Proof. Let (e_n) be the standard basis of ℓ_1 . Choose pairwise distinct projective directions in H ; for instance let

$$\theta_n = \frac{1}{n+1}, \quad h_n = (\cos \theta_n, \sin \theta_n) \in S_H.$$

Define

$$u = \sum_{n=1}^{\infty} 2^{-n} e_n \otimes h_n \quad \text{in } \ell_1 \widehat{\otimes}_\pi H.$$

We first record the standard identification

$$\ell_1 \widehat{\otimes}_\pi H \cong \ell_1(H), \quad \sum_k x_k \otimes y_k \mapsto \left(\sum_k x_k(n) y_k \right)_{n=1}^{\infty},$$

isometrically. Indeed, every finite representation gives

$$\sum_n \left\| \sum_k x_k(n) y_k \right\|_H \leq \sum_n \sum_k |x_k(n)| \|y_k\|_H = \sum_k \|x_k\|_1 \|y_k\|_H,$$

and the reverse inequality follows by representing finite truncations as $\sum_{n \leq N} e_n \otimes v_n$ and passing to the limit in $\ell_1(H)$. Thus the above u corresponds to the sequence $(2^{-n}h_n)_n$, and

$$\|u\|_\pi = \sum_{n=1}^{\infty} 2^{-n} = 1.$$

The displayed countable series for u has sum of product norms equal to 1, so $u \in \text{NA}_\pi(\ell_1 \widehat{\otimes}_\pi H)$.

The tensor u is algebraic. If f_1, f_2 are the coordinate vectors of H , then

$$u = a \otimes f_1 + b \otimes f_2, \quad a = (2^{-n} \cos \theta_n)_n \in \ell_1, \quad b = (2^{-n} \sin \theta_n)_n \in \ell_1.$$

Hence $u \in \ell_1 \otimes H$.

It remains to show $u \notin \text{FNA}_\pi(\ell_1 \widehat{\otimes}_\pi H)$. Suppose, toward a contradiction, that

$$u = \sum_{k=1}^m x_k \otimes y_k \quad \text{and} \quad \sum_{k=1}^m \|x_k\|_1 \|y_k\|_H = \|u\|_\pi = 1.$$

Ignoring zero y_k 's, the n -th coordinate identity is

$$2^{-n}h_n = \sum_{k=1}^m x_k(n)y_k.$$

Therefore

$$2^{-n} = \left\| \sum_{k=1}^m x_k(n)y_k \right\|_H \leq \sum_{k=1}^m |x_k(n)| \|y_k\|_H.$$

Summing over n gives

$$1 \leq \sum_n \sum_k |x_k(n)| \|y_k\|_H = \sum_k \|x_k\|_1 \|y_k\|_H = 1.$$

All nonnegative coordinate deficits are therefore zero. Thus equality holds in the Euclidean triangle inequality for every n :

$$\left\| \sum_{k=1}^m x_k(n)y_k \right\|_H = \sum_{k=1}^m |x_k(n)| \|y_k\|_H.$$

Equality in a strictly convex normed space implies that all nonzero vectors $x_k(n)y_k$ are nonnegative scalar multiples of the same vector. Since their sum is $2^{-n}h_n \neq 0$, at least one summand is nonzero, and every nonzero summand lies on the line $\mathbb{R}h_n$. In particular, for each n , the line $\mathbb{R}h_n$ must be one of the finitely many lines $\mathbb{R}y_1, \dots, \mathbb{R}y_m$.

This is impossible because the directions $\mathbb{R}h_n$ are pairwise distinct. Hence u has no finite optimal representation, so $u \notin \text{FNA}_\pi(\ell_1 \widehat{\otimes}_\pi H)$. $\square$

Remark 1. The same construction works over the complex field with $H = \mathbb{C}^2$, by choosing infinitely many distinct complex one-dimensional subspaces. The equality case in the complex Hilbert-space triangle inequality again forces all nonzero coordinate summands to lie in the same complex line.

Verification and novelty checks

The proof is deterministic and does not rely on numerical computation. The only functional-analytic inputs are the standard isometric identification $\ell_1 \widehat{\otimes}_\pi H = \ell_1(H)$ and equality in the triangle inequality for a strictly convex norm.

Local duplicate checks on June 18, 2026 found no existing solution packet for arXiv:2602.23796 in the run indexes. A prior attempt file for this arXiv id had inspected the same finite-rank question but did not notice this ℓ_1 -coordinate obstruction. Web searches for the exact title of the source paper, for FNA_π , and for “finitely norm-attaining tensors” found the source arXiv record and background material but no later paper answering this specific equality question.

Human-review recommendation

The packet should be reviewed as a likely valid counterexample. The main point to check is the equality-chain argument: a finite optimal representation in $\ell_1(H)$ must be coordinatewise optimal, and strict convexity of H then permits only finitely many coordinate directions.

References

- [1] R. J. Aliaga, S. Dantas, J. Guerrero-Viu, M. Jung, and O. Roldan, *Integral representations of projective norm-attaining tensors*, arXiv:2602.23796, 2026.